SADAT HOSSAIN

AI Research Engineer, Seoul, South Korea
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Objective

Dedicated and experienced Artificial Intelligence Research Engineer with a strong foundation in computer vision, deep learning, and edge computing. Passionate about driving innovation through advanced research and practical applications of AI technologies. Seeksing to contribute my technical expertise, problem-solving skills, and commitment to excellence to impactful projects in a forward-thinking, technology-driven organization.

Skills

Programming Languages: Python, C, C++, Java.

Libraries and Frameworks: Open3D, OpenGL, TensorFlow, Keras, Scikit-learn, H2O, OpenCV, NumPy, Pandas, Matplotlib, PyTesseract, Seaborn, PyTorch, Langchain, LangGraph, Streamlit, ONNX

Tools & Development Environments: IntelliJ, Eclipse, PyCharm, Anaconda, Jupyter Notebook, Android Studio, Git, Docker, Confluence, Jira, Carla Simulator, AI Studio SDK, ChromaDB.

Databases & Data Processing: MongoDB, MySQL, PowerBI, Matlab.

Operating Systems: Linux, Ubuntu, MacOS, Windows.

Web Technologies: XML, HTML, CSS.

Experience

Artificial Intelligence Research Engineer/DeltaX, Seoul, South Korea Feb. 2023 - Present

- Worked on the development of a smart in-cabin monitoring system (ICMS) for autonomous vehicles.
- Utilized facial landmark, gaze estimation, segmentation, classification and detection algorithms along with computer vision techniques for driving monitoring system (DMS) and occupancy monitoring system (OMS).
- Developed and deployed cutting-edge semantic segmentation and object detection models for autonomous vehicles on edge devices (TDA4VM) from Texas Instruments.
- Created a pseudo-LiDAR solution with a mono camera for accurate depth perception in autonomous navigation.
- Applied computer vision techniques and deep learning algorithms for real-time surround view systems in autonomous vehicles.

Graduate Research Assistant/Multimedia Information Processing Lab,
Gwangju, South Korea Mar. 2021 - Feb. 2023

- Improved skin lesion classification and segmentation by leveraging state-of-the-art deep learning techniques.
- Addressed paired clean and noisy images, and video scarcity by employing attention mechanisms, GAN architectures, and Vision Transformers.
- Worked on old image and video denoising by using state-of-the-art deep learning techniques.

Junior Software Developer/Hivecore Limited, Dhaka, Bangladesh Oct. 2019 - Jan. 2021

- Analyzed medical datasets to uncover correlations and trends, utilizing PowerBI for advanced reporting.
- Implemented Python scripts to automate repetitive data analysis tasks, increasing efficiency and reducing manual errors.

Education

MS in Information and Communication Engineering, Chosun University, Gwangju, South Korea.
Mar. 2021 - Dec. 2022

CGPA: 4.31/4.50

Courses include: Theory of Artificial Intelligence, Advanced Artificial Intelligence, and Big Data Management.

BSc in Computer Science & Engineering, North South University, Dhaka, Bangladesh.
Sep. 2014 - Oct. 2018
CGPA: 3.26/4.0

Courses Include: Artificial Intelligence, Fuzzy Logics, Very Large-Scale Integration, Introduction to C, Introduction to Java programming.

Junior Year Project: *Smart Surveillance System [Raspberry Pi, OpenCV]*

Senior Year Project: *A Python based Portable Virtual Text Reader [Tensorflow, Tkinter]*

Relevant Projects

Calibration and Development of pseudo-LiDAR Solution

- Calibrated a mono camera and LiDAR to ensure precise alignment and functionality.
- Utilized a deep learning model to generate a depth map, creating a point cloud from the mono camera.
- Developed a pseudo-LiDAR solution capable of operating at 10 FPS.
- Conducted comparative analysis between pseudo-LiDAR and actual LiDAR solutions, evaluating their performance and effectiveness.

Image and Video Denoising (Funded by SK Telecom, South Korea)

- Generated images and videos with ancient image artifacts using Generative Adversarial Network (GAN) and Vision Transformers.
- Worked on state-of-the-art deep learning models for image and video denoising and noise generation.
- Generated paired dataset to train state-of-the-art denoisers using GAN and vision transformers.

Developed Character-Level Language Model for Bangla Stories using nanoGPT

- Implemented a character-level generative pre-trained transformer (GPT) model following Andrej Karpathy's nanoGPT framework, adapted for Bangla script.
- Utilized PyTorch for model development, leveraging autoregressive language modeling techniques for character-level text generation.
- Integrated multi-headed self-attention and positional encoding to enhance the model's understanding of context and sequence in text.

Analyzing and creating sales reports using MySQL and PowerBI

- Analyzed and pre-processed sales database using Pandas library in python.
- Analyzed and cleaned sales databases using MySQL.
- Perform ETL, create and publish dashboard using PowerBI.

Patent

Title: 이미지및비디오노이즈제거및이미지품질향상을위한쌍별데이터세트생성방법

Inventor: Sadat Hossain

Co-inventors: Nokap Park, Bumshik Lee.

Status: Applied for patent filing, pending review and publication.

Publications

"NG-GAN: A Robust Noise-Generation Generative Adversarial Network for Generating Old-Image Noise"

Journal paper published in Sensors (MDPI), Impact Factor: 3.847, [2024]

Keywords: Deep Learning, Generative Adversarial Network, Noisy Image Generation.

"NTIRE 2024 Challenge on Image Super-Resolution (×4): Methods and Results"

Participated at the Image Restoration and Enhancement workshop in conjunction with CVPR 2024.

"The Second Monocular Depth Estimation Challenge"

Presented at CVPRW 2023, 2nd Monocular Depth Estimation Challenge.

"A Python based Portable Virtual Text Reader"

Presented at the 4th International Conference on Advances in Computing, Communication & Automation (ICACCA-2018), Malaysia.

"Crime Prediction Using Multiple ANFIS Architecture and Spatiotemporal Data"

Presented at the 9th International Conference on Intelligent Systems (IEEEIS-2018), Madeira Island, Portugal.

Keywords: Adaptive Neuro-Fuzzy Inference System, GIS, Spatiotemporal Data, Takagi-Sugeno Fuzzy Model.

Activities & Awards

IEEE Power & Energy Society North South University Student Branch Chapter, Best Volunteer Lead, 2017.

International Conference on Computer and Information Technology (ICCIT), North South University, Bangladesh, Best Volunteer, 2016.